■ Week 2 · Task 4

Linux Basics + Log Files (/var/log)

Internship Practical Report

1. Introduction

This report documents practical Linux and log-analysis learning completed from a SOC analyst perspective. The work focused on command-line navigation, exploring /var/log, viewing logs, inspecting recent events, searching log data, and understanding how logs support cybersecurity investigations.

2. Task Objectives

Practice ls, cd, cat, tail, and grep; explore /var/log; understand failed-login searches and tail -f; complete the required Linux and log-analysis learning; and document practical evidence with screenshots.

3. Linux and Log Analysis Learning

The project includes evidence from Linux Fundamentals Part 1 and Intro to Logs. These learning activities established the Linux and log-analysis concepts used throughout the practical work.

4. Exploring Linux and /var/log

Linux navigation commands such as pwd, ls, and cd were practiced. The /var/log directory was explored because it is a major location for Linux system and application logs.

5. Viewing and Inspecting Logs

The cat command was used to view log-file contents, while tail was used to inspect the newest entries. The command tail -f is used for live monitoring as new log events are written.

6. Searching Logs

grep is used to filter large log files and locate relevant events. A common authentication investigation example is grep 'Failed' auth.log, which searches for failed-login events when that log file and those events are available.

7. SOC Relevance

Logs are security evidence. Analysts use them to investigate authentication activity, service events, errors, suspicious behaviour, and timelines. Parsing, normalisation, enrichment, and correlation help transform raw records into useful investigation data.

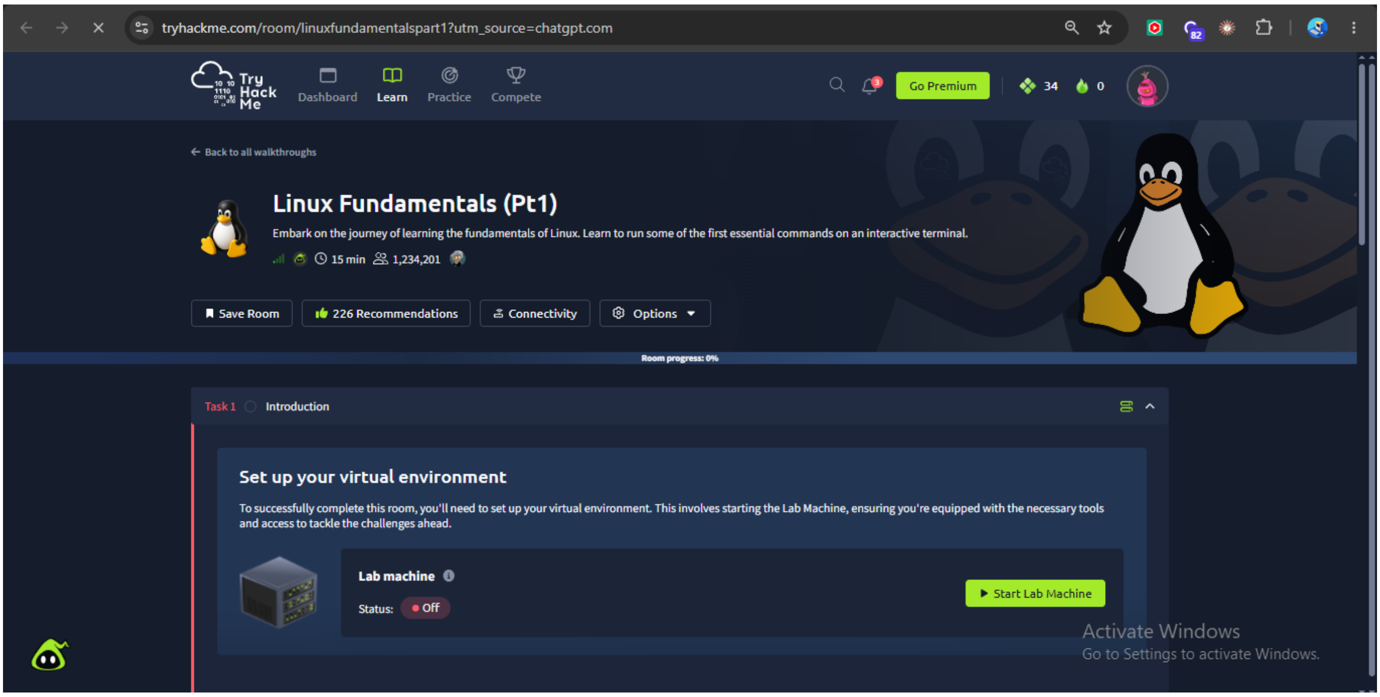
8. Conclusion

This task strengthened foundational Linux and log-analysis skills that support later work with SIEM platforms, centralised logging, threat hunting, and incident response.

■ Screenshot Evidence

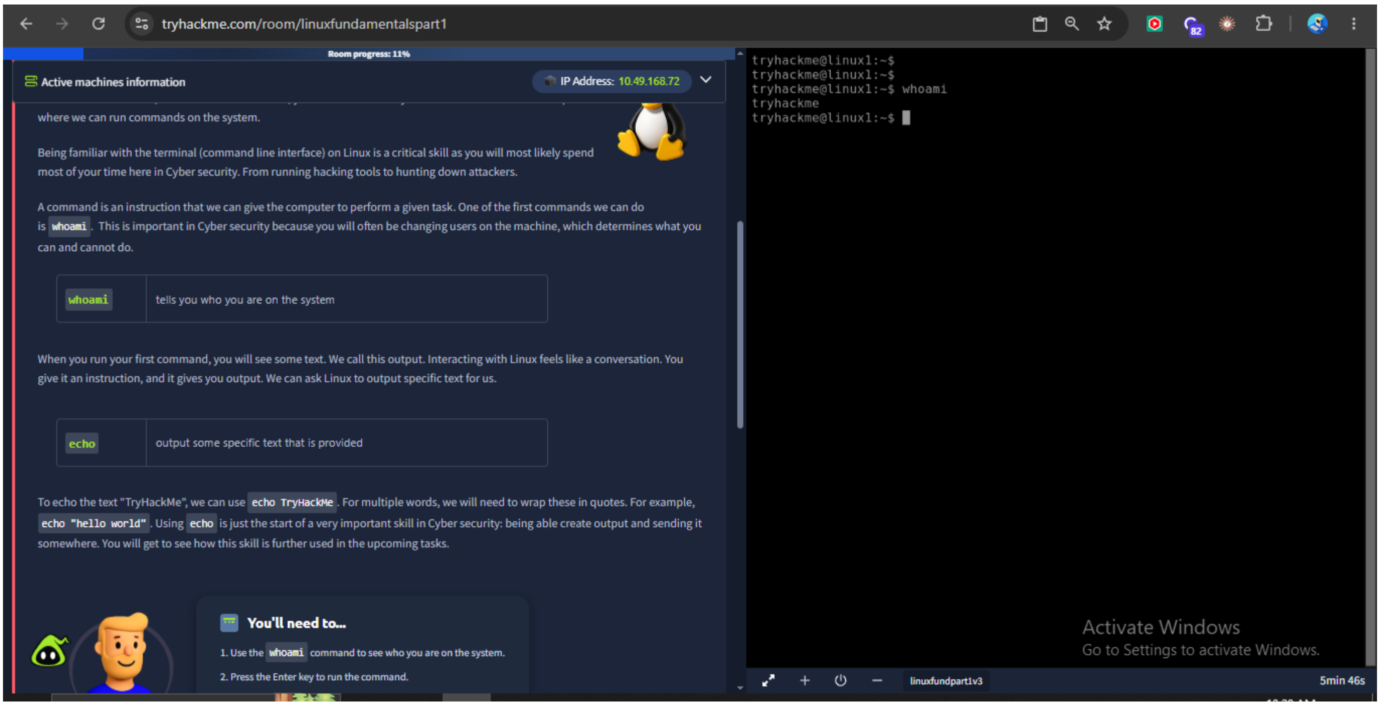
The following pages contain the original screenshot files supplied for this project. Filenames have been preserved.

Screenshot Evidence — l1.png



Screenshot Evidence — l2.png

Screenshot Evidence — l3.png



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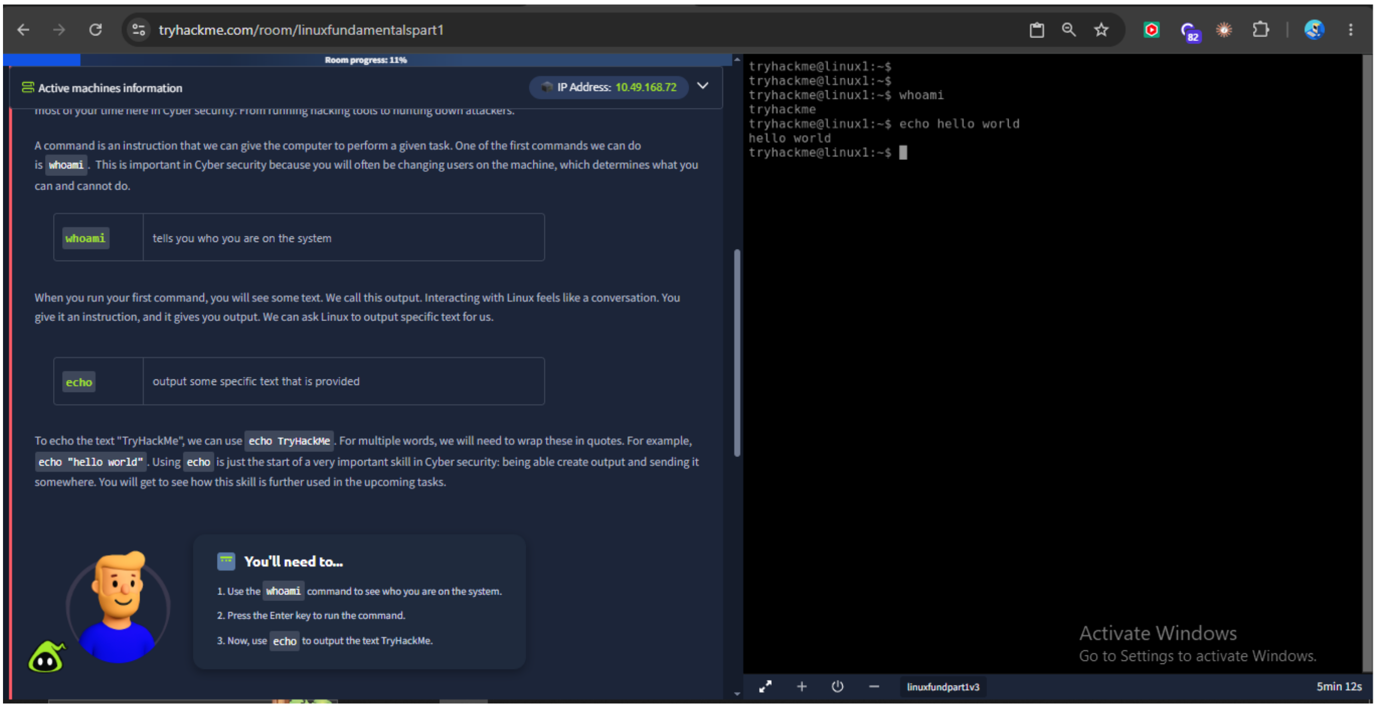

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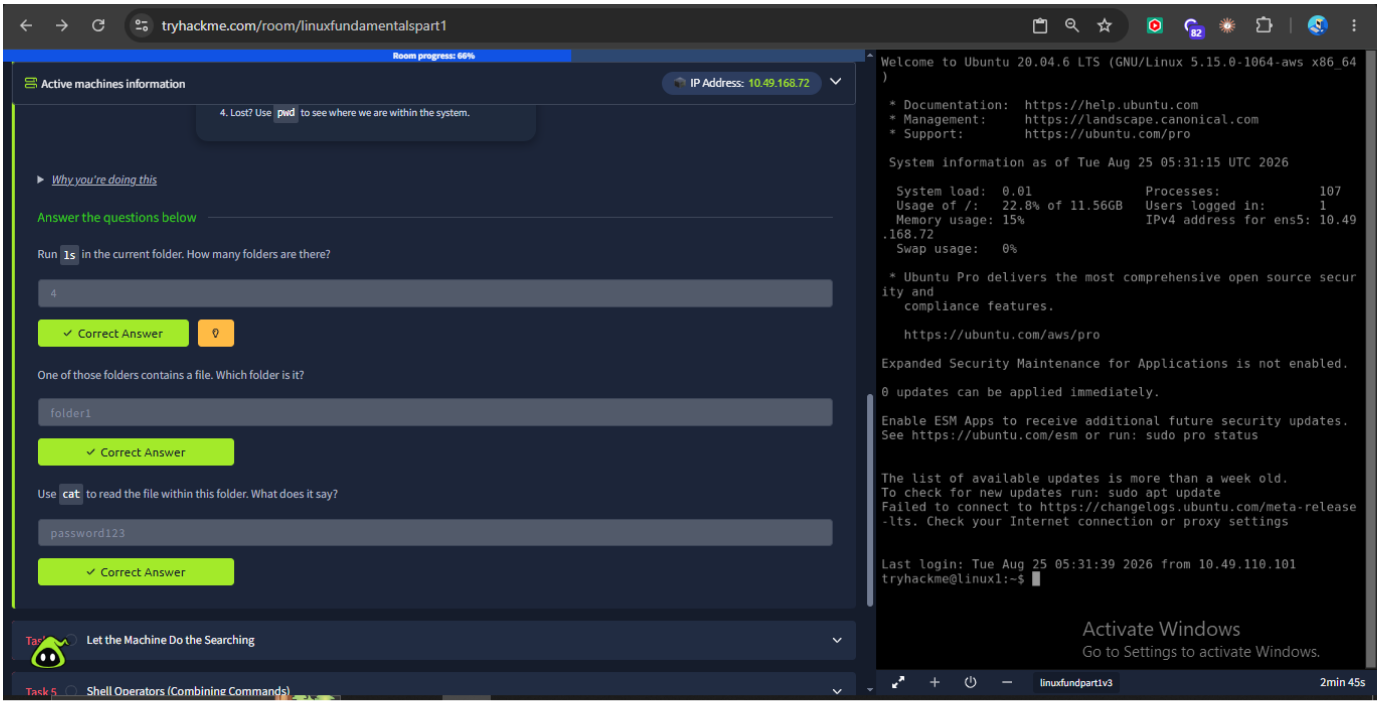

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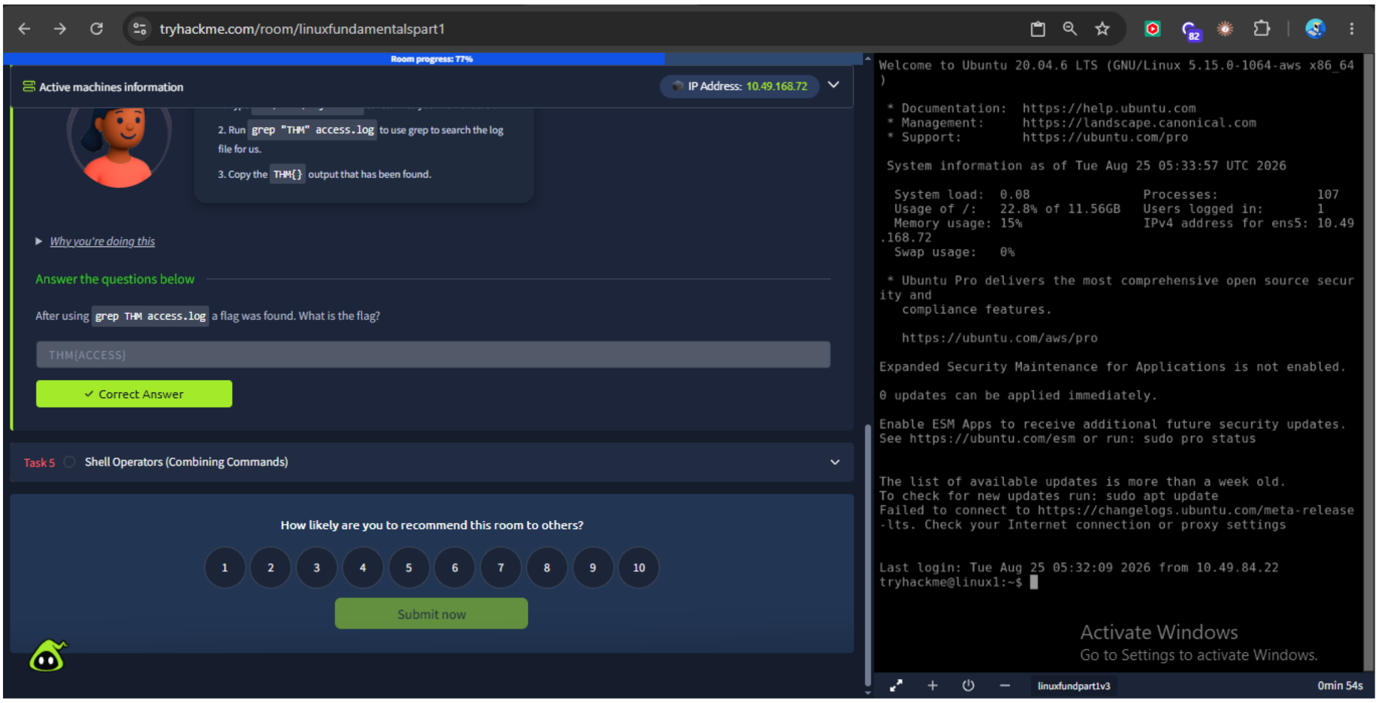

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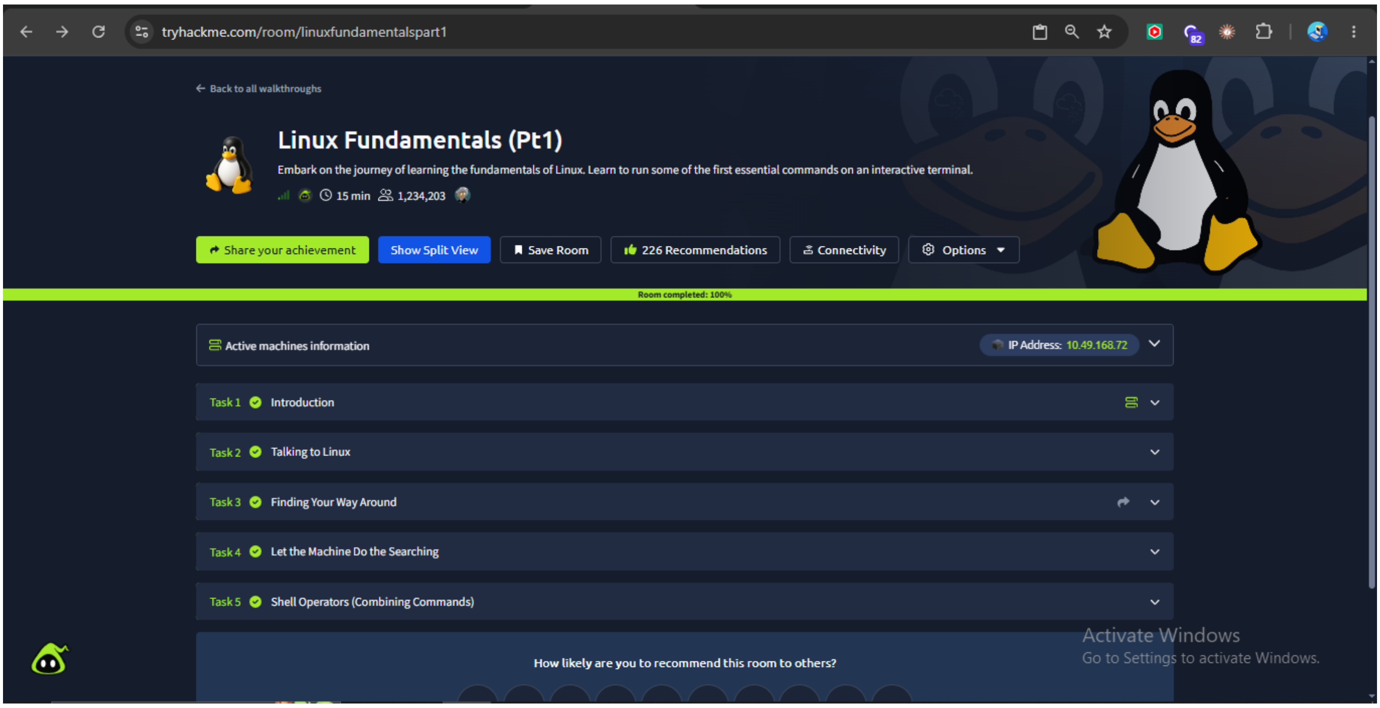

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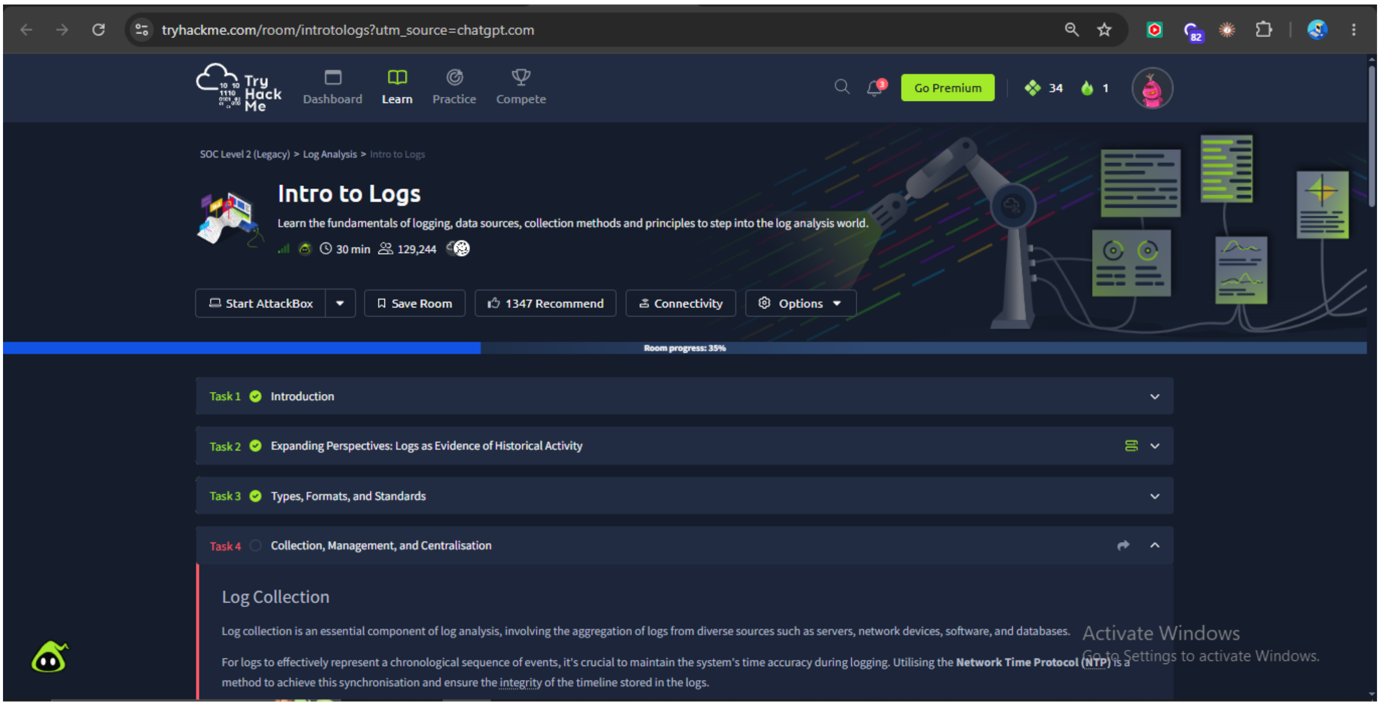

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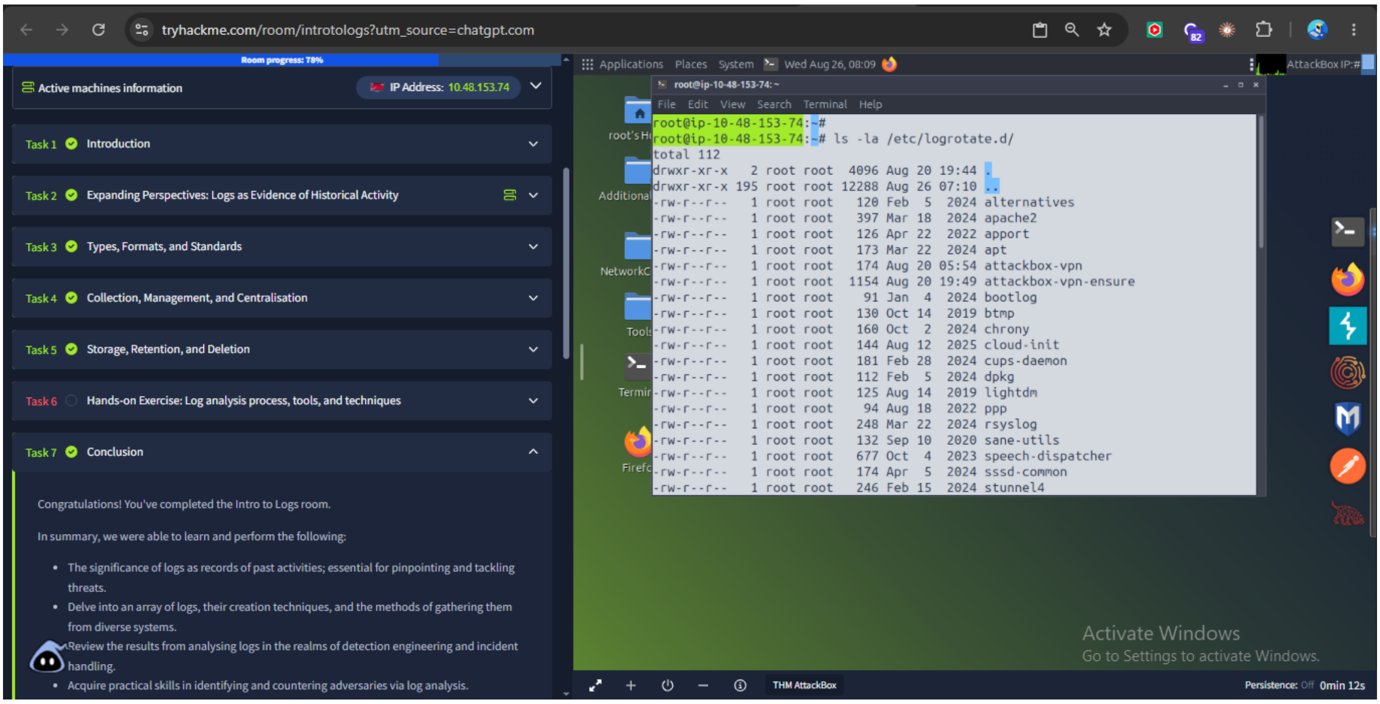
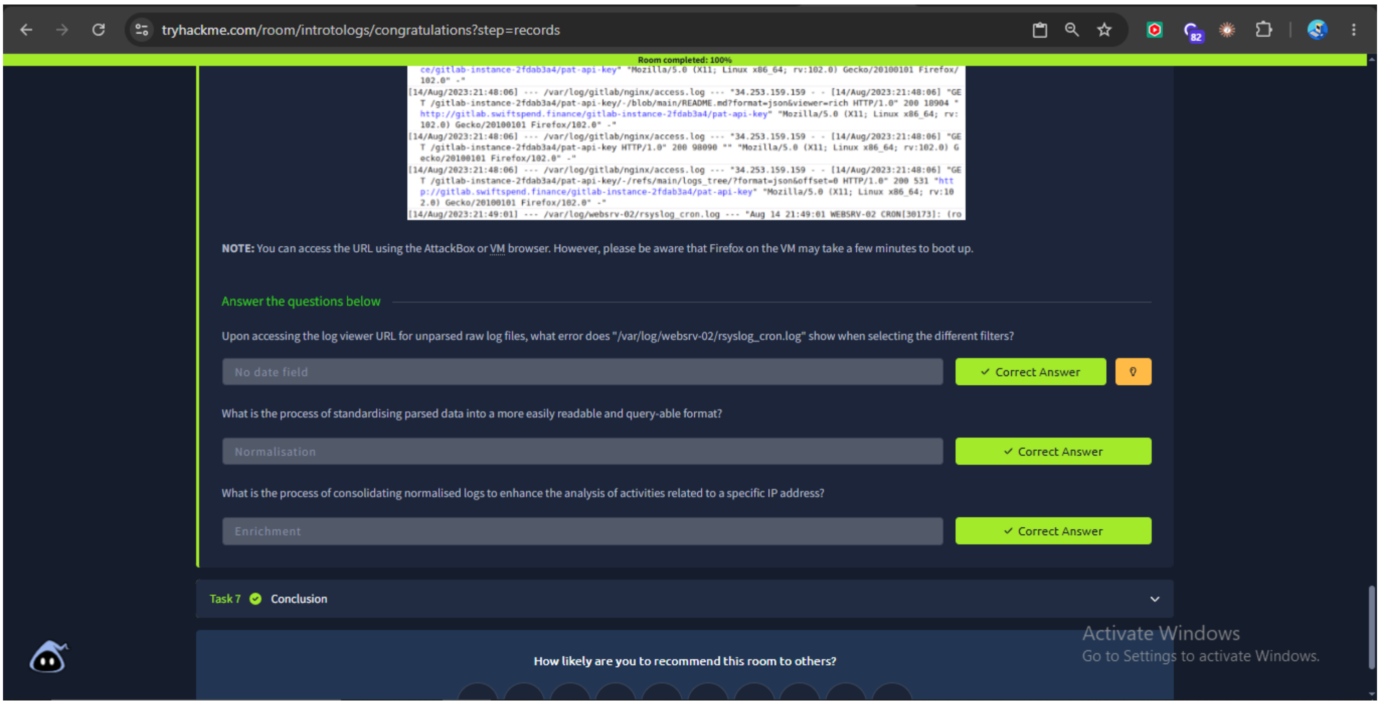


Screenshot Evidence — l10.png



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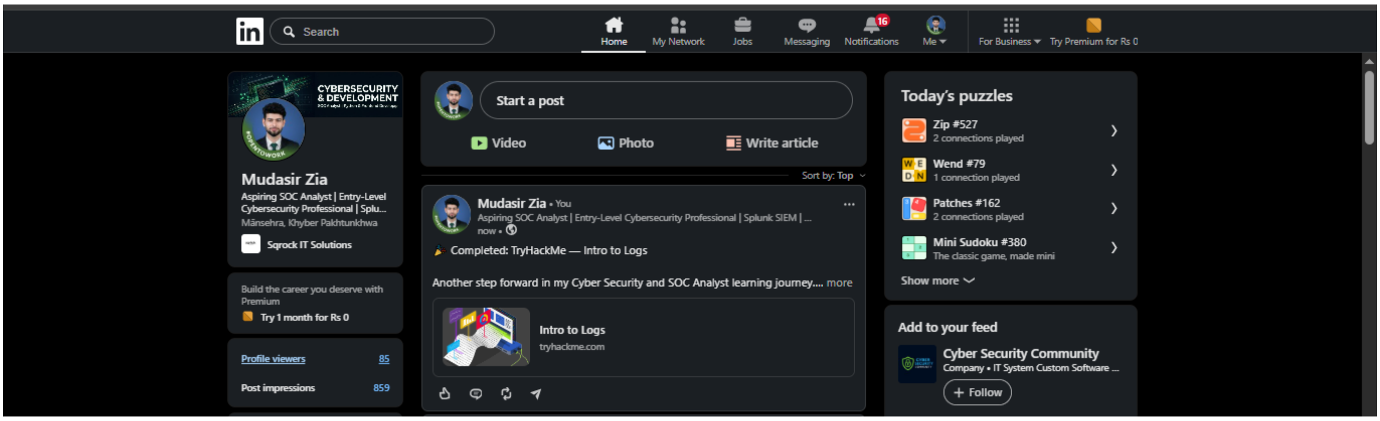

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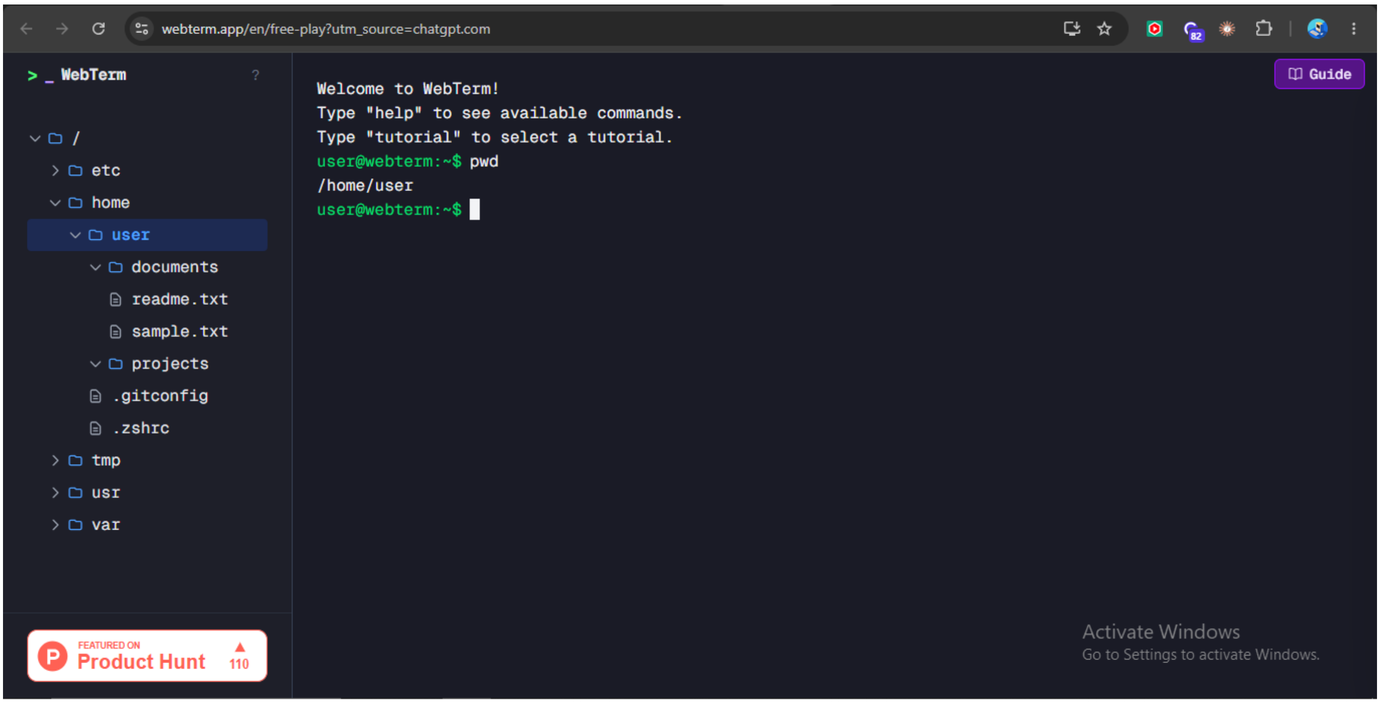

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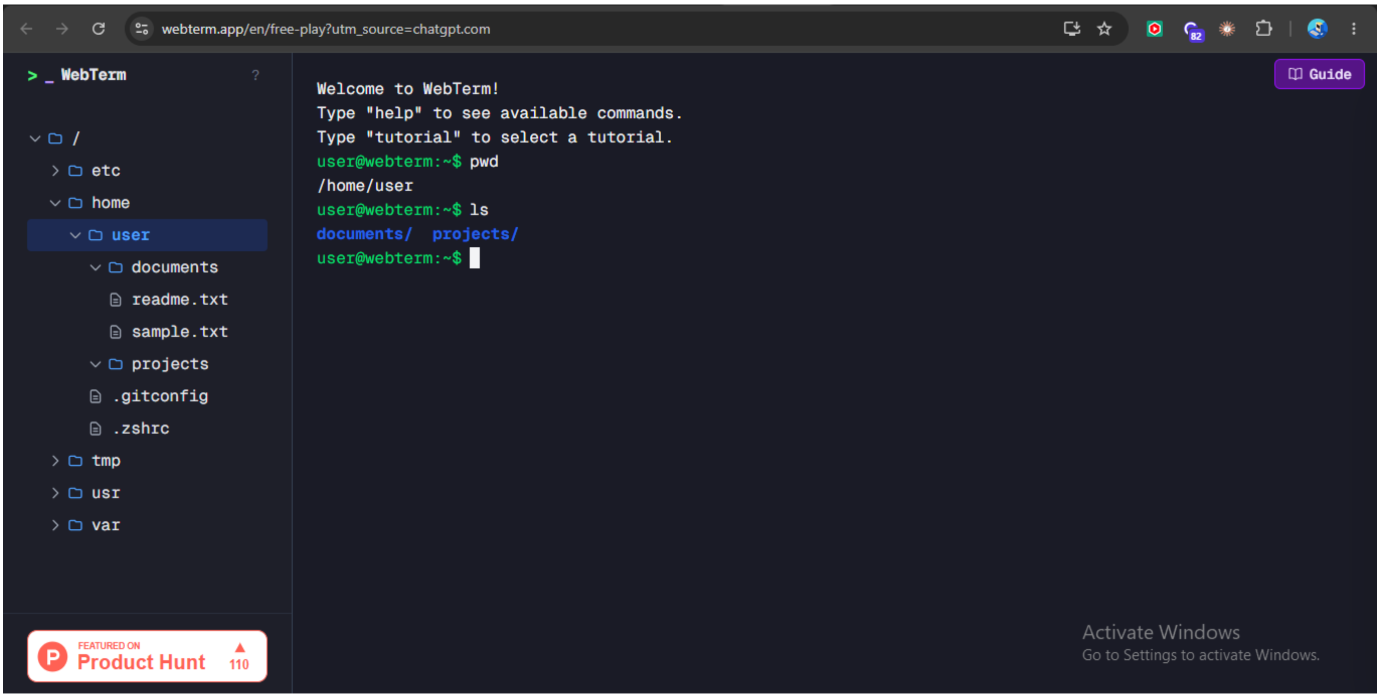

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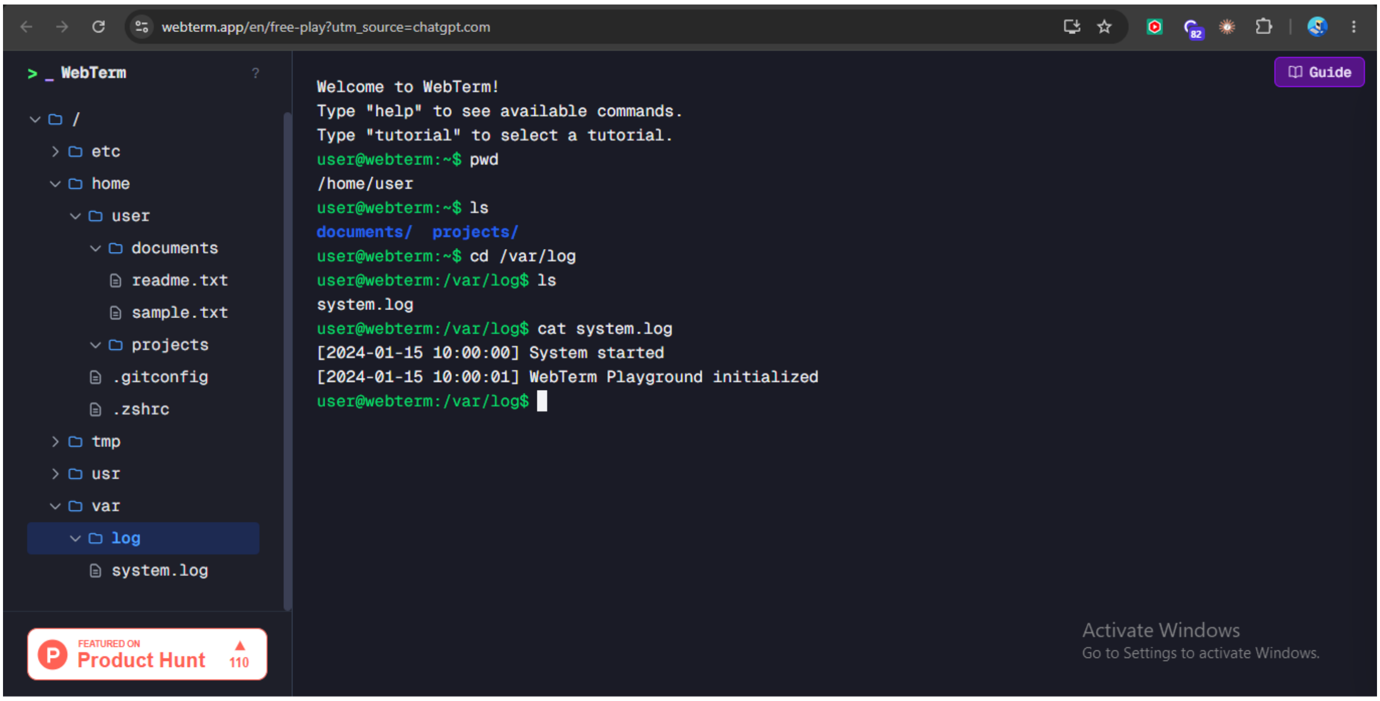

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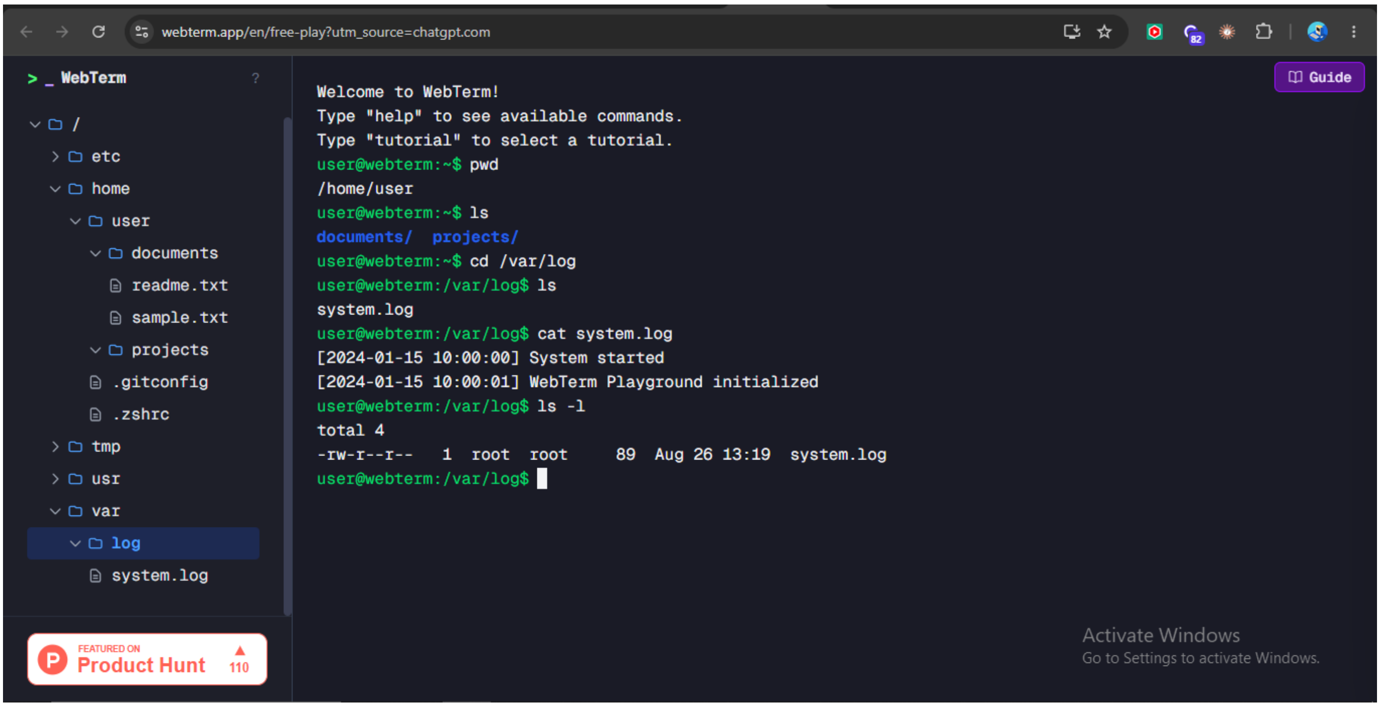

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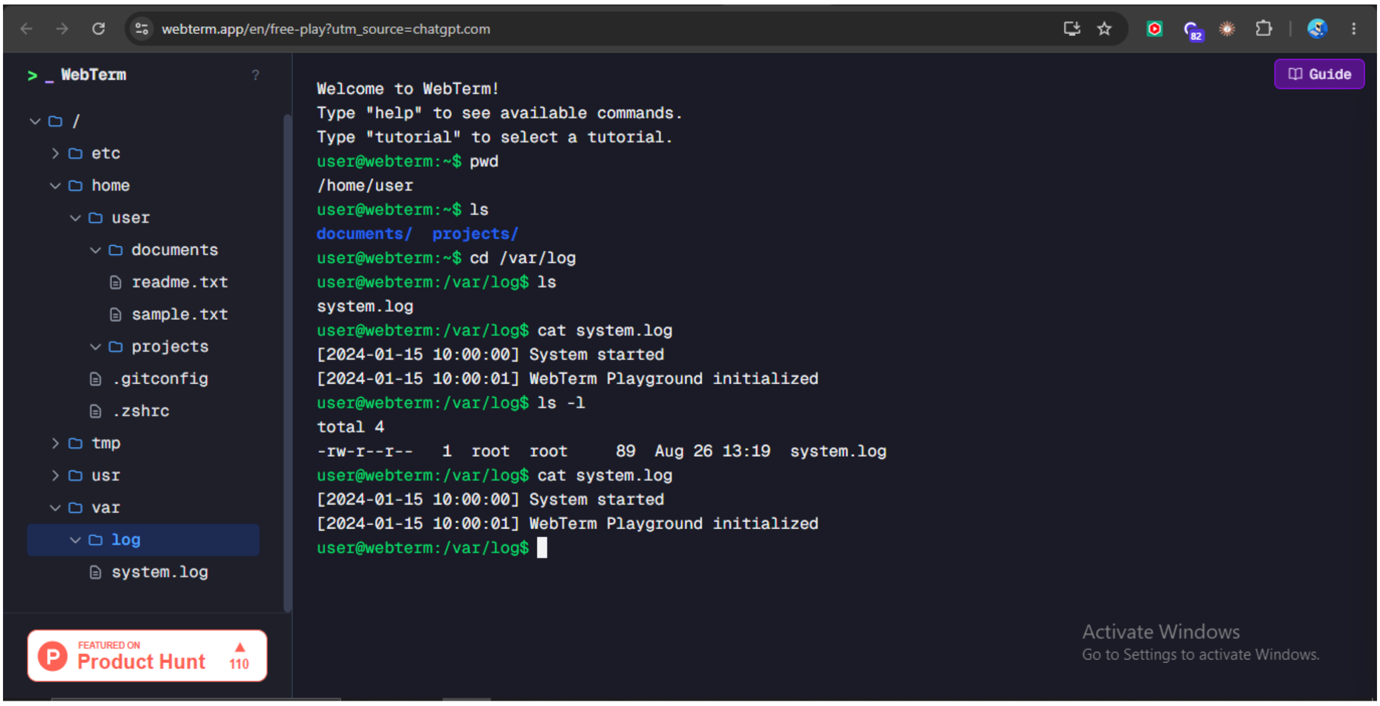

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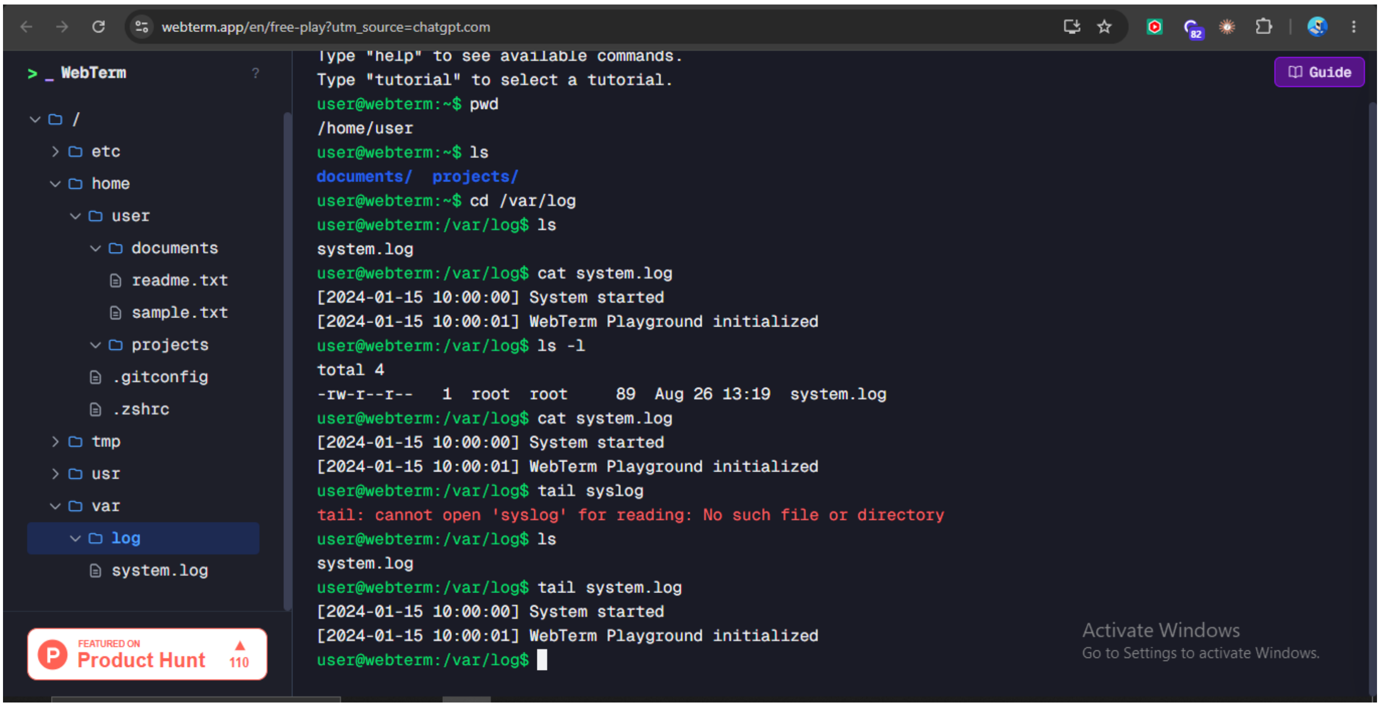

Screenshot Evidence — l18.png